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Improved outcome after severe head injury with a new therapy based on volume regulation and preserved microcirculation

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Abstract

Objective: To assess the new "Lund therapy" of posttraumatic brain edema, based on principles for brain-volume regulation and improved microcirculation.

Design: A prospective, nonrandomized outcome study over a 5-yr period on severely head-injured patients with increased intracranial pressure, comparing the results with a historical control group with the same selection criteria for patients who were treated according to conventional principles.

Setting: General intensive care unit of a university hospital.

Patients: Fifty-three consecutive head-injured patients with a Glasgow Coma Score of <8, and with increased intracranial pressure (>25 mm Hg), despite conventional treatment.

Interventions: Interstitial fluid resorption was obtained by lowering intracapillary hydrostatic pressure, by preserving normal colloid osmotic pressure, and by maintaining a normovolemic (normal albumin/serum and hemoglobin/serum), not overtransfused patient. Intracapillary pressure was reduced by the combination of precapillary vasoconstriction (low-dose thiopental, dihydroergotamine) and reduction of mean arterial pressure, the latter attained with a beta₁-antagonist (metoprolol 0.2 to 0.3 mg/kg/24 hrs iv) and an alpha₂-agonist (clonidine 0.4 to 0.8 [micro sign]g/kg x 4 to 6 iv). Clonidine, in combination with normovolemia, also improves microcirculation by reducing catecholamines in plasma. Intracranial blood volume was reduced by arterial (low-dose thiopental sodium and dihydroergotamine) and large-vein (dihydroergotamine) vasoconstriction. The start dose of dihydroergotamine (maximum 0.9 [micro sign]g/kg/hr) was successively reduced toward discontinuation within 4 to 5 days.

Measurements and Main Results: There were 8% of patients who died and the neurologic conditions of 13% remained severely damaged, compared with 47% and 11%, respectively, for the control group.

Conclusions: The low mortality compared with previous outcome studies strongly indicates that this therapy improves outcome for severe head injuries. However, a randomized, controlled study is needed to reach general acceptance of this new therapy. (Crit Care Med 1998; 26:1881-1886)

Key Words: alpha₂-agonist; beta₁-antagonist; blood-brain barrier; cerebral edema; cerebral microcirculation; cerebral perfusion pressure; cerebral vasoconstriction; head injury; intracranial pressure; outcome

The mortality rate has remained high after severe head trauma, despite extensive research and improved general intensive care. In head-injured patients, mortality and morbidity are closely related to increased intracranial pressure (ICP) [1,2]. Expected mortality has been reported to be as follows: 18% in patients with severe head injury (definition according to Jennett et al. [3]) and ICP values of <20 mm Hg; 45% in patients with ICP values of >20 mm Hg; 74% for ICP values of >40 mm Hg; and 100% for ICP values of >60 mm Hg [4]. These figures are in agreement with most other older and newer outcome studies, including studies in both children and adults; the figures also agree with data obtained from the Traumatic Coma Data Bank [2,3,5-9]. It is apparent, as also stated by Maas [6], that no breakthrough has occurred in the therapy of severe head injury. The following questions must therefore be raised: Is the bad outcome from head trauma mainly a consequence of the severity of the primary injury, which cannot be prevented whatever therapy is given, or are there missing links in the traditional therapeutic approaches?

One main and generally accepted principle in the traditional therapy is to keep a high cerebral perfusion pressure (CPP) to counteract hypoxia and cell damage in areas with compromised blood flow, thereby reducing cellular swelling and increased ICP. Several authors [9-11] have recommended that CPP should be kept >70 to 80 mm Hg, if necessary with the aid of vasoconstrictor therapy or positive inotropic support. However, a high perfusion pressure may simultaneously have the adverse effect of triggering the development of vasogenic brain edema through forces striving toward the classical Starling fluid equilibrium causing an increase in ICP. This increase in ICP will counteract the desired increase in CPP [12] and the patient will come closer to the state of herniation. Transcapillary leakage will occur in areas with disrupted blood-brain barrier. This leakage will be reinforced when the autoregulatory capacity is disturbed since the increased arterial pressure is to a large extent transferred to the capillaries. In these areas, the net fluxes of fluid are determined by the differences in hydrostatic and colloid osmotic pressures across the capillary wall. The use of beta-stimulating inotropic support may also augment the increase in brain edema through beta₂-mediated cerebral vasodilation. Moreover, the use of arterial vasoconstrictors will further enhance what is already present due to the trauma, namely, a strong circulatory vasoconstrictor stress response with high plasma concentrations of catecholamines. This action may impair microcirculation in compromised areas in the brain, as well as in other vascular beds of the body, such as the splanchnic circulation; this result occurs despite an increase in arterial pressure.

Although effective in the short run, the various types of conventional therapy of increased ICP after a head trauma can be criticized as being ineffective, or even harmful, in the long run. The reduction in ICP obtained by controlled hyperventilation thus is short lasting [13] and usually absent in the most severely injured patients with impaired cerebral CO₂ reactivity [14,15]. Moreover, adverse hypoxic effects of pronounced hyperventilation have also been reported [16]. The beneficial effects of repeated infusions of mannitol are ambiguous [17], due to the risk of rebound transcapillary osmotic and hydrostatic effects

[12,18], and no reduction in ICP was obtained in patients with impaired cerebrovascular autoregulation [19] or depressed CO₂ reactivity (unpublished observations). High-dose thiopental is associated with severe pulmonary and other complications [20]; this treatment decreases ICP only in patients with preserved cerebral vasoreactivity [15]. Moreover, when high-dose thiopental was tested in randomized studies, no improvement in outcome was found [21]. These considerations may explain, at least in part, the very high mortality rate in outcome studies reported thus far [2,4,5].

A new therapeutic approach to reduce increased ICP (denoted the "Lund therapy" for severe head injury) was recently introduced by us, together with the first clinical outcome results [22]. The Lund therapy is based on purely physiologic principles for cerebral tissue and blood volume regulation in a tissue with disrupted blood-brain barrier and allows for both the risks of increased ICP and the risks of compromised microcirculation. The therapy thus aims at preventing cerebral hypoxia, simultaneously with taking measures that counteract transcapillary filtration. In summary, our treatment protocol includes the following: a) preservation of a normal colloidal-absorbing force; b) a reduction in intracapillary pressure through a reduction in systemic blood pressure by antihypertensive therapy (a beta₁-antagonist, metoprolol, combined with an alpha₂-agonist, clonidine); and c) a simultaneous, moderate constriction of precapillary resistance vessels with low-dose thiopental and dihydroergotamine. Dihydroergotamine is known to preferentially constrict the large venous vessels [23,24]; it reduces intracranial blood volume mainly through this mechanism. Clonidine in combination with the thiopental and other sedatives and analgesics simultaneously reduces the high-level stress response and the endogenous catecholamine release, which, in turn, improves the microcirculation of the brain as well as the circulation within other tissues. Optimum general intensive care is carried out: fluid therapy to maintain normovolemia (to optimize cerebral microcirculation); monitoring of lung function; nutrition; and electrolyte supplementation.

The present study summarizes our clinical outcome results from a consecutive series of severely head-injured patients with increased ICP, treated according to our new principles during a 5-yr period. Our data show that the clinical outcome in this group of patients is considerably better compared with the outcome derived from published patient data using the same entry criteria but treated according to conventional intensive care principles that include high-dose thiopental, hyperventilation, and preservation of a high CPP [25]. Our clinical outcome is also better than outcomes reported in the literature so far.

MATERIALS AND METHODS

The Department of Neurosurgery, Lund University Hospital, is the only neurosurgical department in the southern medical region in Sweden, and serves a population of about 1.6 million. The department admits 60 to 80 patients annually with severe traumatic brain lesions (definition according to Jennett et al [3]). Focal intracranial mass lesions are evacuated rapidly and continuous recording of ICP is performed in all patients who receive a score of ≥ 4 on the Reaction Level Scale [26], corresponding to a sum score of < 8 on the Glasgow Coma Scale [27]. ICP is recorded via either an intraventricular catheter or an intraparenchymatous microtransducer. The present study included patients (15% to 20%) who were < 65 yrs in age and who developed a dangerous increase in ICP (> 25 mm Hg), despite surgical evacuation of focal mass lesions, adequate sedation, controlled ventilation, and good general intensive care. These criteria are also the entry criteria for the patient to be transferred from the intermediate neurointensive care unit to the general intensive care unit (ICU). Four patients with a manifest brainstem herniation on arrival at the hospital, and who died shortly thereafter (< 8 hrs), were not included in the

study. Two surviving patients with large posttraumatic subarachnoid bleedings were not included in the study, as they were given nimodipine and otherwise were also treated according to conventional principles.

From February 1982 to March 1986, 38 patients were managed in our ICU according to the principles for treating brain edema in use in our ICU at that time, which are similar to the therapy accepted in most neurotrauma centers today. The main therapy to reduce ICP in these patients was controlled hyperventilation and high-dose thiopental, in combination with preservation of a high CPP. This group of patients-the outcome data of which have been presented earlier [20,25]-was used as a reference for the present study, which included 53 patients treated from January 1989 to January 1994. The theoretical principles and the detailed guidelines for our new therapy have been published previously [12,22] and are summarized only in this article.

Reduction of Capillary Hydrostatic Pressure. Mean arterial pressure was reduced to the physiologic level for the age of the individual patient with a combination of the beta₁-antagonist metoprolol (0.2 to 0.3 mg/kg/24 hrs iv) and the alpha (2-agonist) clonidine (0.4 to 0.8 [micro sign]g/kg x 4 to 6 iv) [22,28]. The antihypertensive treatment was initiated when the patients were normovolemic, as obtained by red blood cell and albumin/plasma transfusions to normal albumin and hemoglobin values and to a normal central venous pressure. The antihypertensive treatment always caused a reduction in ICP. A CPP of 60 to 70 mm Hg was considered optimal. However, if it was necessary to control ICP, CPP values of 50 mm Hg for adults and 40 mm Hg for children were accepted. The same zero baseline level was used for blood pressure and ICP monitoring. Dihydroergotamine was given with the purpose of decreasing cerebral blood volume, mainly via constriction of venous vessels. Furthermore, dihydroergotamine, like thiopental, has a precapillary vasoconstrictor effect that contributes to lowering the intracapillary hydrostatic pressure [24,29,30].

Reduction of Cerebral Blood Volume. Intracranial blood volume was reduced, both on the arterial side with thiopental [14] and dihydroergotamine [24,29,30] and on the venous side with dihydroergotamine [23,24]. Dihydroergotamine was always given at the lowest doses necessary to reduce ICP to values of <20 to 25 mm Hg. Dihydroergotamine was not given for >5 days to minimize the risks of compromising peripheral circulation, particularly in patients with fractures of the extremities or renal insufficiency. The maximum doses of dihydroergotamine were: 0.9 [micro sign]g/kg/hr on day 1; 0.6 [micro sign]g/kg/hr on day 2; 0.4 [micro sign]g/kg/hr on day 3; 0.2 [micro sign]g/kg/hr on day 4; and 0.1 [micro sign]g/kg/hr on day 5. In most patients, ICP could be controlled at values of <20 to 25 mm Hg with lower doses and a faster discontinuation of dihydroergotamine than 5 days.

Reduction of Stress Response and Cerebral Energy Metabolism. Stress response was reduced by liberal use of sedatives (benzodiazepines) and analgesics (opioids) already before transfer to the ICU. In the ICU, a further reduction of the stress response and catecholamine release was obtained by a continuous infusion of low-dose thiopental (0.5 to 3 mg/kg/hr) and fentanyl (2 to 5 [micro sign]g/kg/hr), but also by treatment with the beta₁-antagonist metoprolol and the alpha₂-agonist clonidine. The dose of thiopental was low to avoid cardiac inhibition, pulmonary complications, and other side effects [20].

Fluid Balance and Maintenance of Colloid Osmotic Pressure. A balanced or moderately negative fluid balance is a part of the treatment protocol for the purpose of reducing cerebral interstitial space, and is achieved by diuretics (furosemide) and albumin infusion. Supply of antidiuretic hormone analog during

excess urine production was considered contraindicated, unless diabetes insipidus was definitely established (antidiuretic hormone was not given to any patient in the present series). Red blood cell transfusions and albumin were given to achieve normal values (hemoglobin/serum 125 to 140 g/L, albumin/serum [similar]40 g/L) to ensure normovolemia and to optimize oxygen supply. The albumin/plasma/blood transfusions also served the purpose of attaining a normal colloid osmotic pressure favoring transcapillary absorption. All patients were given low-calorie enteral nutrition (maximum energy supply 15 to 20 kcal/kg/24 hrs).

Drainage of Cerebrospinal Fluid. The ventricular catheter could also be used for drainage of cerebrospinal fluid. Except for halting an acute life-threatening increase in ICP, cerebrospinal fluid drainage was not, however, used continuously in the acute phase of increased ICP due to the risk of ventricular collapse and shift of the midline of the brain. In some patients, cerebrospinal fluid drainage was started from a level of ≥ 25 mm Hg in a later phase (>8 days after the trauma) when the blood-brain barrier had started to heal; in such cases, the increased ICP could have been a consequence of inadequate cerebrospinal fluid resorption.

Evaluation of Clinical Outcome. Clinical outcome was evaluated according to the Glasgow Outcome Scale [31] at >6 mos after injury. Clinical outcomes for the reference group were published earlier [25].

Statistical Comparisons. Differences in outcome for patients from the historical reference group treated conventionally and patients given the new therapy were evaluated statistically by chi-square contingency Table analysis with continuity correction.

RESULTS

Of the 53 patients in the present study group, three patients died during ICU treatment and one patient died a few weeks after having been discharged to a hospital abroad. Thus, in all, four patients died within 6 mos of injury. One patient remained in a vegetative state, six patients were classified as being severely disabled, while 42 patients had a favorable outcome (good recovery/moderate disability). Table 1 shows outcome separately in four age groups and four diagnostic groups. Cases denoted as having epidural or subdural hematomas had pure hematomas of the respective category needing surgical evacuation. Intracerebral lesions included focal cerebral contusions and/or intracerebral hematomas evacuated surgically. In some cases, these lesions were combined with an extracerebral hematoma (usually a subdural hematoma). The group "no mass" includes all patients who were not treated with surgical evacuation of focal mass lesions. In the historical control group of 38 patients (1982 to 1986), 18 patients died, two remained in a vegetative state, two were severely disabled, and 16 had a favorable outcome [25].

	Good Recovery, Moderate Disability		Severe Disability Vegetative State		Dead		Total	
	1982-1986	1989-1994	1982-1986	1989-1994	1982-1986	1989-1994	1982-1986	1989-1994
Age Group (yr)								
<21	8	20	0	2	12	1	20	23
21-40	6	15	1	3	5	2	12	20
41-60	2	6	3	2	1	1	6	9
>60	0	1	0	0	0	0	0	1
Sum	16	42	4	7	18	4	38	53
Percent	42	79	11	13	47	8	—	—
Diagnostic Group								
Epidural	1	1	1	0	0	0	2	1
Subdural	1	5	0	1	2	0	3	6
Intracerebral	6	17	3	5	6	2	15	24
No mass	8	19	0	1	10	2	18	22
Sum	16	42	4	7	18	4	38	53
Percent	42	79	11	13	47	8	—	—

Table 1. Outcome according to Glasgow Outcome Scale

Mortality was significantly ($p < .0001$) lower in the group treated according to the new protocol than in the group treated according to conventional principles. The ratio of patients with outcomes denoted as vegetative or severe disability was about the same. The ratio of patients with a favorable outcome (good recovery/moderate disability) thus was significantly ($p < .001$) higher in the group treated according to the new protocol.

DISCUSSION

This study shows that mortality was considerably reduced in head-injured patients not developing a manifest brainstem herniation or dying within the first few hours after arrival at the hospital, when treated according to the specific principles presented above. The main principles are based on the physiologic mechanisms controlling the volume in a tissue with depressed autoregulation and with capillaries permeable for solutes [12,22,32], while microcirculation is simultaneously preserved or improved. Comparing the outcome results with those results reported so far in the literature [2,4-9], the present study shows that conventional therapies as a whole are much less effective, which may indicate that there are missing links in the conventional therapeutic approaches.

Nonrandomized studies using historical control groups always suffer from limitations, with biases such as the following: different timing of patient entry into the study; varying degrees of doctor and nurse engagement; and varying quality of general intensive care at different times. Therefore, a randomized, controlled study is preferable. However, when obtaining such a marked improvement in outcome in pathophysiologically defined groups of patients, the use of a historical control group should be relevant under the following conditions: a) the two groups are selected for ICU treatment according to the same principles, with no significant differences in distribution of age or type of lesion; and b) the two groups are treated in the same ICU. These conditions were true in this study. The fact that the outcome results in our historical control group were similar to those results reported in most of the other older and newer head-injury outcome studies also supports the relevance of the present study.

Our new treatment protocol includes two concurrent physiologic principles: a) a rapid decrease in ICP is obtained by a reduction in cerebral blood volume by the vasoconstrictive effects of dihydroergotamine and thiopental; and b) a gradual and lasting decrease in ICP is obtained by increasing transcapillary fluid

absorption by reducing intracapillary hydrostatic pressure and by maintaining a normal colloid osmotic pressure. The treatment protocol thus comprises two controversial components. First, the arterial part of the reduction in cerebral blood volume might be expected to decrease cerebral blood flow, which would increase the risks of cerebral ischemia. Second, the reduction in intracapillary hydrostatic pressure is obtained by a combination of precapillary vasoconstriction and a reduction in CPP, which might further increase the risks of cerebral ischemia. Arguments will be given below, however, that the risk to induce hypoxia with the vasoconstrictor therapy used is small; moreover, antihypertensive therapy, when used in combination with a balanced fluid substitution, will not impair cerebral oxygenation.

We agree with the common view that insults secondary to ischemia significantly contribute to bad outcome [6]. Our therapy, however, is based on the assumption that the most important pathophysiologic consequences of traumatic brain injury are not only neuronal and astrocytic cell swelling, but also the release of inflammatory substances, which cause the opening of the blood-brain barrier and result in increased fluid filtration and raised ICP [33-35]. This view supports the importance of evacuating contusions and preserving the microcirculation around contused areas. The latter objective is accomplished in our therapy by maintaining normovolemia, normal hemoglobin and albumin concentrations, inhibiting catecholamine release by clonidine [36], using sedatives, and maintaining an optimum fluid balance.

There is always a risk associated with the use of vasoconstricting drugs in the treatment of traumatic brain injury, as these drugs may induce hypoxia. However, the arterial vasoconstrictor effect of thiopental is related to, and is a direct consequence of, its metabolism-inhibiting effect; therefore, thiopental will not induce hypoxia. Providing that the recommended doses are not exceeded, the relative risk of hypoxia is also small during the infusion of dihydroergotamine compared with other vasoconstrictor mechanisms sometimes used to reduce ICP (e.g., hyperventilation [15,16], phenylephrine [9], norepinephrine [9], and indomethacin [37]), as the main part of the reduction in blood volume occurs on the venous side [23,24]. This understanding finds support in a previous study [29] on severely head-injured patients (with preserved, as well as impaired, cerebral CO₂ reactivity), in which a bolus infusion of dihydroergotamine (4.0 [micro sign]g/kg) significantly decreased ICP but only slightly increased cerebral vascular resistance, allowing cerebral blood flow and cerebral arterial-venous oxygen difference to remain unchanged. Single photon emission computed tomography measurements confirmed a simultaneous marked reduction in intracranial blood volume. In vitro studies of human blood vessels confirmed that dihydroergotamine at low concentrations causes a more pronounced contraction of small cortical veins than cortical arteries [24].

Indomethacin has been suggested as an alternative vasoconstrictor to reduce intracranial blood volume during treatment of increased ICP in severe head injuries [37]. However, indomethacin is a potent constrictor of cerebral precapillary vessels [38], and the reduction in ICP in severe head injuries is obtained at the expense of a significant decrease in cerebral blood flow and an increase in cerebral arterial-venous oxygen difference [37]. An experimental study in piglets [39] with induced intracranial hypertension showed that indomethacin caused a significant decrease in ICP only when cerebral blood flow, cerebral arterial-venous oxygen difference, and electroencephalogram indicated imminent cerebral ischemia.

Since dihydroergotamine is a rather nonspecific drug, affecting various 5-hydroxytryptamine (5-HT), dopamine-2, and alpha₁- and alpha₂-adrenergic receptors [40], a more specific cerebral (venous) vasoconstrictor might be preferred. Sumatriptan is generally regarded as an agonist selective for the

5-HT₁-like receptor subtype [41] and was used for reduction of increased ICP in an experimental study in piglets [42]. In these studies, intermediate doses of sumatriptan (0.03 and 0.1 mg/kg) reduced ICP but with a simultaneous significant decrease in cerebral blood flow, and the electroencephalographic pattern changed with significant decreases in fast frequencies and increases in slow activity. At a higher dose of sumatriptan (0.5 mg/kg), ICP instead increased, probably due to dilation of the intracranial large veins [42]. Thus, compared with dihydroergotamine, both sumatriptan and indomethacin seem to be less suited for the treatment of increased ICP. So far, no clinical studies have been performed with sumatriptan.

The desired reduction in intracapillary hydrostatic pressure is primarily obtained by a pharmacologically induced decrease in mean arterial pressure, with a combination of an alpha₂-agonist (clonidine) and a beta₁-antagonist (metoprolol). Antihypertensive vasodilator drugs causing cerebral vasodilation were not used, as these drugs may increase hydrostatic intracapillary pressure causing increased transcapillary filtration. The antihypertensive effect of clonidine is the net effect of the reduced sympathetic outflow from the brain and a direct peripheral alpha-mediated vasoconstrictor effect [36]. However, the direct effects of alpha stimulation on cerebral circulation are known to be small compared with the alpha effects in most other tissues [43], and no direct influence on cerebral circulation was obtained when the alpha₂-agonist clonidine was given to patients with severe traumatic brain lesions [28]. Although clonidine may induce an adverse, direct, alpha-mediated cerebral vasoconstrictor effect in areas with compromised circulation, most likely this effect is more than counteracted by the clonidine-induced inhibition of endogenously released catecholamines [36]. The beta₁-adrenergic influence on cerebral vasculature has been shown to be small under physiologic conditions [44]. In patients with severe traumatic brain lesions, no direct cerebral hemodynamic effects were obtained after infusion of the beta₁-antagonist metoprolol (0.1 mg/kg) [28]. The decrease in ICP always seen after the start of the antihypertensive therapy also contributes to preservation of the CPP, despite a lower arterial inflow pressure. Thus, there are factors inherent in our antihypertensive therapy that counteracted hypoxia. This observation is supported by the unchanged or increased cerebral venous bulb saturation values found after start of the therapy, as measured in some of our patients.

According to our protocol, the antihypertensive drugs are given only when the patient is normovolemic and when all surgical mass lesions have been evacuated. The lowest accepted CPP was 50 mm Hg for adults and 40 mm Hg for children. Even lower values were sometimes accepted, but only during shorter periods of high ICP values. Usually, the lowest CPP was 55 to 65 mm Hg, and increased slowly during the first 36 hrs of treatment due to a progressive decrease in ICP [22]. Presently, the optimum CPP in head-injured patients is controversial, and most authors [10,11] advocated that the CPP should be kept at >70 to 80 mm Hg. The present study shows that excellent clinical results can be obtained with an active reduction in arterial inflow pressure, while providing a simultaneous, effective stress- and catecholamine-reducing therapy, a normovolemic state, and optimum intensive care. Preliminary data indicate that cerebral biochemical monitoring of extravascular metabolites reflecting the energy state, with the use of bedside microdialysis from multiple intracerebral catheters during intensive care, gives valuable information for optimum treatment. This technique was recently introduced in our ICU, and is presently used for all patients treated according to our protocol for evaluation of the "Lund therapy" [45].

The present study shows that mortality after a severe head injury can be reduced considerably with a therapy using physiologic principles for brain-volume regulation and preserved microcirculation. The main objective would be to restrict ICP to reasonable levels and simultaneously preserve an adequate blood flow supply. Although this study indicates that the "Lund therapy" is superior to conventional

treatments of head injuries, the study suffers from the limitations that are inherent in nonrandomized studies. Therefore, there is a recognized need for a randomized, controlled study, whereby traditional medical therapy is compared with our new therapy.

REFERENCES

1. Miller JD, Becker DP, Ward JD, et al: Significance of intracranial hypertension in severe head injury. J Neurosurg 1977; 47:503-516 [\[Context Link\]](#)
2. Marmarou A, Anderson RL, Ward JD, et al: Impact of ICP instability and hypotension on outcome in patients with severe head trauma. Report on The Traumatic Coma Bank Data. J Neurosurg 1991; 75(Suppl):59-66 [\[Context Link\]](#)
3. Jennett B, Teasdale G, Galbraith S, et al: Severe head injuries in three countries. J Neurol Neurosurg Psychiatry 1977; 40:291-298 [\[Context Link\]](#)
4. Miller JD: Significance and management of intracranial hypertension in head injury. In: Intracranial Pressure V. Ishii S, Nagai H, Brock M (Eds.). Berlin, Heidelberg, Springer-Verlag, 1983, pp 44-53 [\[Context Link\]](#)
5. Marshall LF, Gattille T, Klauger MR, et al: The outcome of severe closed head injury. J Neurosurg 1991; 75(Suppl):S28-S36 [\[Context Link\]](#)
6. Maas AIR: Pathophysiology, monitoring and treatment of severe head injury. In: Yearbook of Intensive Care and Emergency Medicine. Vincent JL (Ed). Berlin, Heidelberg, Springer-Verlag, 1993, pp 555-578 [\[Context Link\]](#)
7. Berger MS, Pitts LH, Lovely M, et al: Outcome from severe head injury in children and adolescents. J Neurosurg 1985; 62:194-199 [\[Context Link\]](#)
8. Pigula FA, Wald SL, Shackford SR, et al: The effect of hypotension and hypoxia on children with severe head injuries. J Pediatr Surg 1993; 28:310-316 [\[Context Link\]](#)
9. Rosner MJ, Rosner SD, Johnson AH: Cerebral perfusion pressure: Management protocol and clinical results. J Neurosurg 1995; 83:949-962 [\[Context Link\]](#)
10. Miller JD, Dearden NM, Piper IR, et al: Control of intracranial pressure in patients with severe head injury. J Neurotrauma 1992; 9(Suppl 1):317-326 [\[Context Link\]](#)

11. Rosner MJ, Daughton S: Cerebral perfusion pressure management in head injury. *J Trauma* 1990; 30:933-941 [\[Context Link\]](#)

12. Grande PO, Asgeirsson B, Nordstrom CH: Physiological principles for volume regulation of a tissue enclosed in a rigid shelter with application to the injured brain. *J Trauma* 1997; 42(Suppl):23-31 [\[Context Link\]](#)

13. Christensen MS: Acid-base changes in cerebrospinal fluid and blood, and blood volume changes following prolonged hyperventilation in man. *Br J Anaesth* 1974; 46:348-357 [\[Context Link\]](#)

14. Nordstrom C-H, Messeter K, Sundbarg G, et al. Cerebral blood flow, vasoreactivity and oxygen consumption during barbiturate therapy in severe traumatic brain lesions. *J Neurosurg* 1988; 68:424-431 [\[Context Link\]](#)

15. Schalen W, Messeter K, Nordstrom C-H: Cerebral vasoreactivity and the prediction of outcome in severe traumatic brain lesions. *Acta Anaesthesiol Scand* 1991; 35:113-122 [\[Context Link\]](#)

16. Muizelaar JP, Marmarou A, Ward JD, et al: Adverse effects of prolonged hyperventilation in patients with severe head injury: A randomised clinical trial. *J Neurosurg* 1991; 75:731-739 [\[Context Link\]](#)

17. Kullberg G, Sundbarg G: Reduction of raised intracranial pressure following infusion of mannitol. A review of clinical pressure recordings. In: *Intracranial Pressure III*. Becks JWF, Bosch DA, Brock M (Eds). Berlin, Heidelberg, Springer-Verlag, 1976, pp 224-230 [\[Context Link\]](#)

18. Kaufmann AM, Cardoso ER: Aggravation of vasogenic cerebral edema by multiple dose mannitol. *J Neurosurg* 1992; 77:584-589 [\[Context Link\]](#)

19. Muizelaar JP, Lutz HA, Becker DP: Effect of mannitol on ICP and CBF and correlation with pressure autoregulation in severely head-injured patients. *J Neurosurg* 1984; 61:700-706 [\[Context Link\]](#)

20. Schalen W, Messeter K, Nordstrom C-H: Complications and side effects during thiopentone therapy in patients with severe head injuries. *Acta Anaesthesiol Scand* 1992; 36:369-377 [\[Context Link\]](#)

21. Ward JD, Becker DP, Miller JD, et al: Failure of prophylactic barbiturate coma in the treatment of severe head injury. *J Neurosurg* 1985; 62:383-388 [\[Context Link\]](#)

22. Asgeirsson B, Grande PO, Nordstrom C-H: A new therapy of post-trauma brain oedema based on haemodynamic principles for brain volume regulation. *Intensive Care Med* 1994; 20:260-267 [[Context Link](#)]
23. Mellander S, Nordenfelt I: The effects of dihydroergotamine and noradrenaline on resistance, exchange and capacitance function in the peripheral circulation. *Clin Science* 1970; 39:183-201 [[Context Link](#)]
24. Nilsson F, Nilsson T, Edvinsson L, et al.: Effects of dihydroergotamine and sumatriptan on isolated human cerebral and peripheral arteries and veins. *Acta Anaesthesiol Scand* 1997; 41:1257-1262 [[Context Link](#)]
25. Schalen W, Sonesson B, Messeter K, et al: Clinical outcome and cognitive impairment in patients with severe head injuries treated with barbiturate coma. *Acta Neurochir* 1992; 117:153-159 [[Context Link](#)]
26. Starmark JE, Stalhammar D, Holmgren E: The Reaction Level Scale (RLS 85). Manual and Guidelines. *Acta Neurochir* 1988; 91:12-20 [[Context Link](#)]
27. Teasdale G, Jennett B: Assessment of coma and impaired consciousness. A practical scale. *Lancet* 1974; ii:81-84 [[Context Link](#)]
28. Asgeirsson B, Grande PO, Nordstrom C-H, et al: Effects of hypotensive treatment with alpha₂-agonist and beta₁-antagonist on cerebral haemodynamics in severe head injury. *Acta Anaesthesiol Scand* 1995; 39:347-351 [[Context Link](#)]
29. Asgeirsson B, Grande PO, Nordstrom C-H, et al: Cerebral haemodynamic effects of dihydroergotamine in patients with severe traumatic brain lesions. *Acta Anaesthesiol Scand* 1995; 39:922-930 [[Context Link](#)]
30. Nilsson F, Messeter K, Grande P-O, et al: Effects of dihydroergotamine on cerebral circulation during experimental intracranial hypertension. *Acta Anaesthesiol Scand* 1995; 39:916-921 [[Context Link](#)]
31. Jennett B, Bond M: Assessment of outcome after severe brain damage. A practical scale. *Lancet* 1975; i:480-484 [[Context Link](#)]
32. Asgeirsson B, Grande PO: Effects of arterial and venous pressure alterations on transcapillary fluid exchange during raised tissue pressure. *Intensive Care Med* 1994; 20:567-572 [[Context Link](#)]

33. Holmin S, Mathiesen T, Shetye J, et al: Intracerebral inflammatory response to experimental brain contusion. *Acta Neurochir* 1995; 132:110-119 [\[Context Link\]](#)

34. Homin S, Mathiesen T: Biphasic edema development after experimental brain contusion. *Neurosci Lett* 1995; 194:97-100 [\[Context Link\]](#)

35. Holmin S, Schalling M, Hojeberg B, et al: Delayed cytokine expression in rat brain following experimental contusion. *J Neurosurg* 1996; 86:493-504 [\[Context Link\]](#)

36. Payen D, Quintin L, Plaisence P, et al: Head injury: Clonidine decreases plasma catecholamines. *Crit Care Med* 1990; 18:392-395 **Buy Now** | [\[Context Link\]](#)

37. Jensen K, Ohrstrom J, Cold GE, et al: The effects of indomethacin on intracranial pressure, cerebral blood flow and cerebral metabolism in patients with severe head injury and intracranial hypertension. *Acta Neurochir (Wien)* 1991; 108:116-121 [\[Context Link\]](#)

38. Pickard JD, MacKenzie ET: Inhibition of prostaglandin synthesis and the response of baboon cerebral circulation to carbon dioxide. *Nature (New Biol)* 1973; 245:187-188 [\[Context Link\]](#)

39. Nilsson F, Bjorkman S, Rosen I, et al: Cerebral vasoconstriction by indomethacin in intracranial hypertension. An experimental investigation in pigs. *Anesthesiology* 1995; 83:1283-1992 **Ovid Full Text** | [\[Context Link\]](#)

40. McCarthy BG, Peroutka SJ: Comparative neuropharmacology of dihydroergotamine and sumatriptan (GR 43175). *Headache* 1989; 29:420-422 [\[Context Link\]](#)

41. Jansen I, Edvinsson L, Mortensen A, et al: Sumatriptan is a potent vasoconstrictor of human dural arteries via a 5-HT₁-like receptor. *Cephalalgia* 1992; 12:202-205 [\[Context Link\]](#)

42. Nilsson F, Nilsson T, Edvinsson L, et al: Sumatriptan-induced cerebral vasoconstriction as treatment of experimental intracranial hypertension. *Acta Anaesthesiol Scand* 1996; 40:612-620 **Buy Now** | [\[Context Link\]](#)

43. Edvinsson L, MacKenzie ET: Amine mechanisms in the cerebral circulation. *Pharmacol Rev* 1976; 2:275-346 [\[Context Link\]](#)

44. Jensen K, Bunemann I, Riisager S, et al: Cerebral blood flow during anaesthesia: Influence of pretreatment with metoprolol. *Br J Anaesth* 1989; 62:321-323 [\[Context Link\]](#)

45. Ungerstedt U, Backstrom T, Hallstrom A, et al: Microdialysis in normal and injured human brain. In: Physiology, Stress, and Malnutrition. Kinney JM, Tucker H (Eds). Philadelphia, New York, Lippincott-Raven Publishers, 1997, pp 361-374 [\[Context Link\]](#)

IMAGE GALLERY

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	Adult Mammals (Relative Sensitivity)		Neonatal Mammals (Relative Sensitivity)		Total		Total
	1982-1989	1990-1994	1982-1989	1990-1994	1982-1989	1990-1994	
Age Group (yr)							
<25	4	36	0	0	11	7	36
25-49	4	10	1	4	7	1	12
50-69	4	1	0	0	5	0	5
>70	0	1	0	0	0	0	1
Male	10	38	1	4	16	8	38
Female	0	0	0	0	0	0	0
Occupational Category							
Executive	1	1	0	0	0	0	1
Professional	1	1	0	0	0	0	1
Administrative	0	0	0	0	0	0	0
Skilled	0	0	0	0	0	0	0
Unskilled	0	0	0	0	0	0	0
Total	10	38	1	4	16	8	38
Percent	26	38	11	32	44	20	33

☐ Table 1

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